



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0905 JAVA PROGRAMMING – CO4 – AT1

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Q1.

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
import java.util.ArrayList;
```

```
public class FoodOrder extends Frame implements ActionListener {
```

```
    CardLayout card = new CardLayout();
```

```
    Panel cards = new Panel(card);
```

```
    TextField name, table, qty;
```

```
    Choice category, food;
```

```
    TextArea orderArea, billArea;
```

```
    Button add, generate, clear, next, previous;
```

```
    ArrayList<String> items = new ArrayList<>();
```

```
    ArrayList<Integer> prices = new ArrayList<>();
```

```
    ArrayList<Integer> quantities = new ArrayList<>();
```

```
    FoodOrder() {
```

```
        setTitle("Food Ordering and Billing System");
```

```
        setSize(600,500);
```

```
        setLayout(new BorderLayout());
```

```
Panel p1 = new Panel(new GridLayout(4,2,5,5));
p1.add(new Label("Customer Name:"));
name = new TextField();
p1.add(name);
p1.add(new Label("Table Number:"));
table = new TextField();
p1.add(table);
next = new Button("Next");
previous = new Button("Previous");
p1.add(next);
p1.add(previous);
next.addActionListener(this);
previous.addActionListener(this);
cards.add(p1,"customer");
```

```
Panel p2 = new Panel(new BorderLayout());
Panel top = new Panel(new GridLayout(4,2,5,5));
```

```
top.add(new Label("Food Category:"));
category = new Choice();
category.add("Main Course");
category.add("Snacks");
category.add("Drinks");
top.add(category);
```

```
top.add(new Label("Food Item:"));
```

```
food = new Choice();
food.add("Burger - 120");
food.add("Pizza - 250");
food.add("Biryani - 180");
food.add("French Fries - 80");
food.add("Sandwich - 100");
food.add("Coke - 50");
top.add(food);

top.add(new Label("Quantity:"));
qty = new TextField();
top.add(qty);

add = new Button("Add Item");
clear = new Button("Clear");
top.add(add);
top.add(clear);

add.addActionListener(this);
clear.addActionListener(this);
p2.add(top, BorderLayout.NORTH);

orderArea = new TextArea();
p2.add(orderArea, BorderLayout.CENTER);

Panel b = new Panel();
```

```
previous = new Button("Previous");
next = new Button("Next");
b.add(previous);
b.add(next);
previous.addActionListener(this);
next.addActionListener(this);
p2.add(b, BorderLayout.SOUTH);
cards.add(p2, "selection");
```

```
Panel p3 = new Panel(new BorderLayout());
billArea = new TextArea();
p3.add(billArea, BorderLayout.CENTER);
```

```
Panel bb = new Panel();
generate = new Button("Generate Bill");
previous = new Button("Previous");
bb.add(generate);
bb.add(previous);
generate.addActionListener(this);
previous.addActionListener(this);
p3.add(bb, BorderLayout.SOUTH);
cards.add(p3, "bill");
```

```
add(cards, BorderLayout.CENTER);
setVisible(true);
```

```
addWindowListener(new WindowAdapter() {  
    public void windowClosing(WindowEvent e) {  
        System.exit(0);  
    }  
});  
}
```

```
public void actionPerformed(ActionEvent e) {  
    if(e.getSource()==next) card.next(cards);  
    if(e.getSource()==previous) card.previous(cards);
```

```
    if(e.getSource()==add) {  
        if(qty.getText().isEmpty()) {  
            orderArea.append("Enter quantity.\n");  
            return;  
        }  
        try {  
            int q = Integer.parseInt(qty.getText());  
            if(q<=0) {  
                orderArea.append("Invalid quantity.\n");  
                return;  
            }  
        }
```

```
        String item = food.getSelectedItem();  
        int price = 0;
```

```
if(item.contains("Burger")) price=120;
else if(item.contains("Pizza")) price=250;
else if(item.contains("Biryani")) price=180;
else if(item.contains("French Fries")) price=80;
else if(item.contains("Sandwich")) price=100;
else if(item.contains("Coke")) price=50;
```

```
items.add(item);
prices.add(price);
quantities.add(q);
orderArea.append(item+" x "+q+" = Rs."+price*q+"\n");
qty.setText("");
} catch(Exception ex) {
    orderArea.append("Invalid quantity.\n");
}
}
```

```
if(e.getSource()==clear) {
    items.clear();
    prices.clear();
    quantities.clear();
    orderArea.setText("");
    qty.setText("");
}
```

```
if(e.getSource()==generate) {
```

```

double subtotal=0;

billArea.setText("");

billArea.append("FOOD BILL\n");

billArea.append("Customer: "+name.getText()+"\n");

billArea.append("Table: "+table.getText()+"\n");


for(int i=0;i<items.size();i++) {

    double cost=prices.get(i)*quantities.get(i);

    subtotal+=cost;

    billArea.append(items.get(i)+"      x      "+quantities.get(i)+"      =
Rs."+cost+"\n");

}


double gst=subtotal*0.05;

billArea.append("Subtotal: Rs."+subtotal+"\n");

billArea.append("GST 5%: Rs."+gst+"\n");

billArea.append("Final Bill: Rs."+(subtotal+gst)+"\n");

}

}

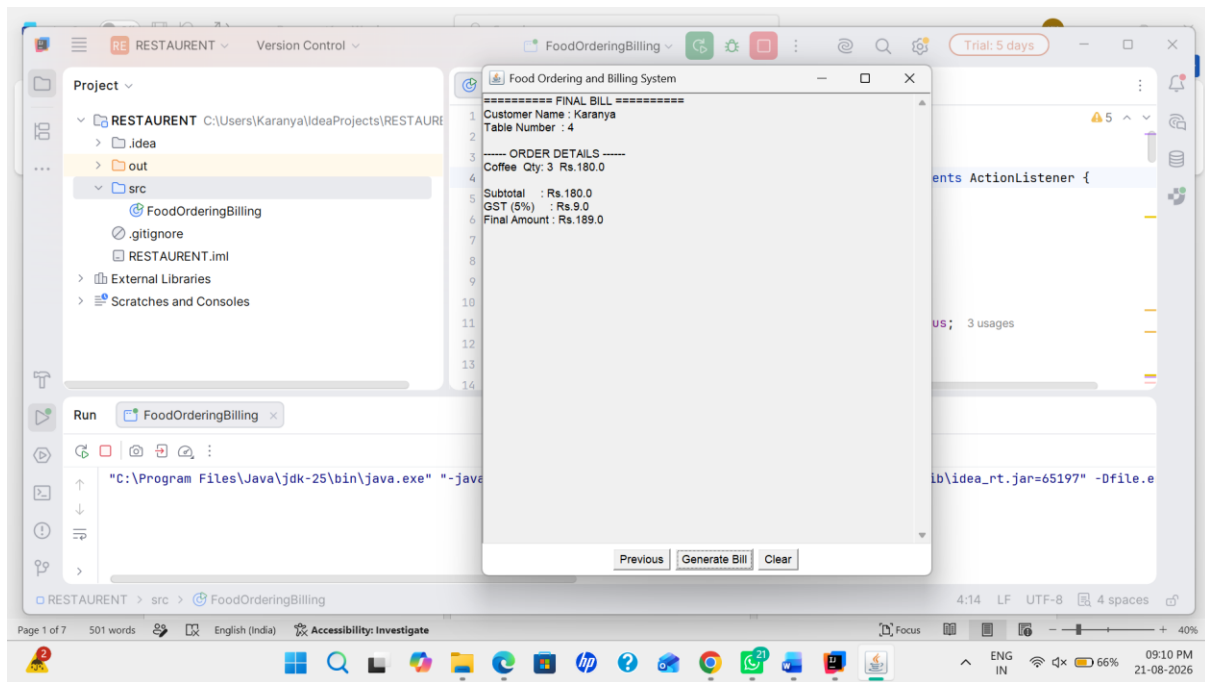

public static void main(String[] args) {

    new FoodOrder();

}

}

```



Q2.

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class FoodOrderingBilling extends Frame implements ActionListener {
```

```
    CardLayout cardLayout;
```

```
    Panel cardPanel;
```

```
    TextField customerName, tableNo;
```

```
    Choice category, foodItem, quantity;
```

```
    List orderList;
```

```
    TextArea billArea;
```

```
    Button addItem, generateBill, clear, next, previous;
```

```
    double subtotal = 0;
```

```
    FoodOrderingBilling() {
```

```
        setTitle("Food Ordering and Billing System");
```

```
        setSize(700, 550);
```

```
        setLayout(new BorderLayout());
```

```
        cardLayout = new CardLayout();
```

```
        cardPanel = new Panel(cardLayout);
```

```
        cardPanel.add(customerCard(), "customer");
```



```

cardPanel.add(orderCard(), "order");
cardPanel.add(billCard(), "bill");
add(cardPanel, BorderLayout.CENTER);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});
setLocationRelativeTo(null);
setVisible(true);
}

```

```

Panel customerCard() {
    Panel panel = new Panel(new GridLayout(4, 2, 10, 10));
    panel.add(new Label("Customer Name:"));
    customerName = new TextField();
    panel.add(customerName);
    panel.add(new Label("Table Number:"));
    tableNo = new TextField();
    panel.add(tableNo);
    panel.add(new Label(""));
    next = new Button("Next");
    next.addActionListener(this);
    panel.add(next);
    return panel;
}

```

```

Panel orderCard() {
    Panel panel = new Panel(new BorderLayout(10, 10));
    Panel form = new Panel(new GridLayout(4, 2, 10, 10));
    category = new Choice();
    category.add("Main Course");
}

```

```
category.add("Snacks");
category.add("Drinks");
foodItem = new Choice();
quantity = new Choice();
quantity.add("1");
quantity.add("2");
quantity.add("3");
quantity.add("4");
quantity.add("5");
updateFoodItems();
category.addItemListener(new ItemListener() {
    public void itemStateChanged(ItemEvent e) {
        updateFoodItems();
    }
});
form.add(new Label("Food Category:"));
form.add(category);
form.add(new Label("Food Item:"));
form.add(foodItem);
form.add(new Label("Quantity:"));
form.add(quantity);
addItem = new Button("Add Item");
addItem.addActionListener(this);
form.add(new Label(""));
form.add(addItem);
panel.add(form, BorderLayout.NORTH);
orderList = new List(10);
panel.add(orderList, BorderLayout.CENTER);
Panel buttons = new Panel();
previous = new Button("Previous");
next = new Button("Next");
previous.addActionListener(this);
```

```
next.addActionListener(this);  
buttons.add(previous);  
buttons.add(next);  
panel.add(buttons, BorderLayout.SOUTH);  
return panel;  
}
```

```
Panel billCard() {  
    Panel panel = new Panel(new BorderLayout());  
    billArea = new TextArea();  
    billArea.setEditable(false);  
    panel.add(billArea, BorderLayout.CENTER);  
    Panel buttons = new Panel();  
    generateBill = new Button("Generate Bill");  
    clear = new Button("Clear");  
    previous = new Button("Previous");  
    generateBill.addActionListener(this);  
    clear.addActionListener(this);  
    previous.addActionListener(this);  
    buttons.add(previous);  
    buttons.add(generateBill);  
    buttons.add(clear);  
    panel.add(buttons, BorderLayout.SOUTH);  
    return panel;  
}
```

```
void updateFoodItems() {  
    foodItem.removeAll();  
    if (category.getSelectedItem().equals("Main Course")) {  
        foodItem.add("Burger");  
        foodItem.add("Pizza");  
        foodItem.add("Fried Rice");  
    }
```

```

    } else if (category.getSelectedItem().equals("Snacks")) {
        foodItem.add("French Fries");
        foodItem.add("Sandwich");
        foodItem.add("Samosa");
    } else {
        foodItem.add("Coffee");
        foodItem.add("Tea");
        foodItem.add("Fresh Juice");
    }
}

```

```

double getPrice(String item) {
    if (item.equals("Burger")) return 120;
    if (item.equals("Pizza")) return 250;
    if (item.equals("Fried Rice")) return 180;
    if (item.equals("French Fries")) return 100;
    if (item.equals("Sandwich")) return 90;
    if (item.equals("Samosa")) return 40;
    if (item.equals("Coffee")) return 60;
    if (item.equals("Tea")) return 30;
    if (item.equals("Fresh Juice")) return 80;
    return 0;
}

```

```

public void actionPerformed(ActionEvent e) {
    String command = e.getActionCommand();

    if (command.equals("Add Item")) {
        String item = foodItem.getSelectedItem();
        int qty = Integer.parseInt(quantity.getSelectedItem());
        double total = getPrice(item) * qty;
        subtotal += total;
    }
}

```

```

        orderList.add(item + " Qty: " + qty + " Rs." + total);
    } else if (command.equals("Next")) {
        cardLayout.next(cardPanel);
    } else if (command.equals("Previous")) {
        cardLayout.previous(cardPanel);
    } else if (command.equals("Generate Bill")) {
        double gst = subtotal * 0.05;
        double finalAmount = subtotal + gst;
        billArea.setText("===== FINAL BILL =====\n");
        billArea.append("Customer Name : " + customerName.getText() + "\n");
        billArea.append("Table Number : " + tableNo.getText() + "\n\n");
        billArea.append("----- ORDER DETAILS ----- \n");
        for (int i = 0; i < orderList.getItemCount(); i++)
            billArea.append(orderList.getItem(i) + "\n");
        billArea.append("\nSubtotal    : Rs." + subtotal);
        billArea.append("\nGST (5%)      : Rs." + gst);
        billArea.append("\nFinal Amount : Rs." + finalAmount);
    } else if (command.equals("Clear")) {
        customerName.setText("");
        tableNo.setText("");
        orderList.removeAll();
        billArea.setText("");
        subtotal = 0;
        cardLayout.first(cardPanel);
    }
}

public static void main(String[] args) {
    new FoodOrderingBilling();
}
}

```

